

ABRA — the plain-English deck

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A slide-by-slide, jargon-light tour. The white paper (linked on the last slide) has the math and sources.

Slide 1 — ABRA

A family of small AI models for competitive Pokémon (Champions VGC). Built from thousands of real ladder games. Honest about what it can and can't know.

Slide 2 — The big idea

You **can't reliably predict who wins** a match from the two team sheets — in this format it's close to a coin flip, and even a strong player-rating model barely beats a coin. So ABRA stops trying to call the winner and instead **helps you make better decisions**: what to bring, and what to do turn by turn.

This is the same move sports analytics made with "expected goals" — you can't predict the final score, but you can measure the value of each shot. ABRA is that idea, pointed at Pokémon.

Slide 3 — How it's built

One rule: **keep every real game, and analyse on top of it**. ABRA saves every public ladder replay into a growing library and builds its models on that library — so it gets smarter as more games come in, and never has to re-download anything. It runs on a normal laptop; no special hardware.

Slide 4 — The town of models

Each model is a "house" you can visit on the site:

- **MEDICHAM** — the damage engine. Knows exactly how hard every move hits. (Checked against the community standard and matches it.)
 - **GURU** — reads the metagame: which team styles beat which, from real results, with error bars.
 - **XATU** — reads the opponent: the likely item, ability, and moves behind each Pokémon.
 - **PORY** — the win meter: mid-battle, tells you your real chance to win.
 - **SLOWKING** — the strategist: there's no single best team, so it plays a smart mix.
 - **CHOMP** — the team picker: which four to bring, which two to lead.
 - **ALAKAZAM** — the coach (in progress): the best move to make, right now.
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Slide 5 — The win

PORY beats a coin flip. Mid-game, using how many Pokémon each side has left and their health, it calls your win chance far better than chance — and it's *calibrated*, meaning when it says 70% it's really about 70%. It's live on the site as a per-turn "you're at X%".

Slide 6 — The honest parts (this is a feature)

Two things we tested and reported straight, even though they're negatives:

- **Picking the team doesn't beat a coin.** We tested whether the team-picker's choices track who wins — they don't, more than chance. So we don't oversell it. The damage math is still exact and useful.
- **The "rock-paper-scissors" metagame is a hint, not a fact.** The data suggests Trick Room beats Hyper Offense beats Sand beats Trick Room — but each of those rests on only ~13–18 games, so the error bars are wide. We label it suggestive, and it'll firm up as more games arrive.

Saying what we *can't* prove, as plainly as what we can, is the whole point.

Slide 7 — What's next

- **ALAKAZAM** — the in-battle coach. A light version runs in your browser; a version that could beat top humans needs a rented cloud computer and months of the AI playing itself to learn.
 - **A million games.** Real games trickle in ~18k/week; the AI playing itself can generate millions fast, which is what the coach needs to get truly strong.
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Slide 8 — Read more

Full technical detail, math, and sources: [ABRA white paper](#). Also: [project summary](#) · [technical docs](#) · [model ledger](#).